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**Факультет «Информатика и системы управления»
Кафедра ИУ5 «Системы обработки информации и управления»**

Курс «ПиКЯП»
Отчет по РК №2

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Листинг кода исправленной РК1:

```
import random

class Section:
    def __init__(self, id, name):
        self.id = id
        self.name = name

class Document:
    def __init__(self, id, section_id, title, size):
        self.id = id
        self.section_id = section_id
        self.title = title
        self.size = size

class DocSection:
    def __init__(self, section_id, doc_id):
        self.section_id = section_id
        self.document_id = doc_id

sections = [Section(i, f"Раздел {i}") for i in range(1, 5)]
sections.append(Section(5, "белиберда без слова на Р"))

documents = [Document(i, random.randint(1, 5), f"Документ {i}",
random.randint(100, 900)) for i in range(1, 8)]

DocSections = [DocSection(random.randint(1, 5), random.randint(1, 7)) for i in
range(8)]

def task1(documents, sections):
    result = []
    for section in sorted(sections, key = lambda s: s.name):
        docs = sorted([doc for doc in documents if doc.section_id == section.id],
key=lambda doc: doc.id)
        result.append((section.name, [doc.title for doc in docs]))
    return result

def task2(documents, sections):
    result = []
    for section in sections:
        docs = [doc for doc in documents if doc.section_id == section.id]
        docSum = sum(doc.size for doc in docs)
        result.append((section.name, docSum))
    result = sorted(result, key=lambda section: section[1], reverse=True)
    return result

def task3(documents, sections, doc_sections):
    result = []

    section_dict = {}
```

```

    for ds in doc_sections:
        section_dict.setdefault(ds.section_id, []).append(ds.document_id)

    for section in sections:
        if section.id in section_dict and "раздел" in section.name.lower():
            docs = [
                doc.title
                for doc in documents
                if doc.id in section_dict[section.id]
            ]
            result.append((section.name, docs))

    return result

def main():
    result1 = task1(documents, sections)
    print("Задание 1: Выведите список всех связанных разделов и документов,
отсортированных по разделам, сортировка по документам произвольная")
    for cort in result1:
        print(cort[0] + ':')
        for doc in cort[1]:
            print('\t' + doc + ',')

    result2 = task2(documents, sections)
    print("Задание 2: Выведите список разделов с суммарным размером документов в
каждом разделе, отсортированный по суммарному размеру.")
    for cort in result2:
        print(cort[0] + ':')
        print('\t' + str(cort[1]))

    result3 = task3(documents, sections, DocSections)
    print("Задание 3: Разделы, содержащие слово 'раздел', и документы в них:")
    for section, docs in result3:
        print(section)
        for doc in docs:
            print("\t", doc)

if __name__ == "__main__":
    main()

```

Листинг кода тестов:

```

from main import task1, task2, task3, Section, Document, DocSection

sections = [Section(i, f"Раздел {i}") for i in range(1, 5)]
sections.append(Section(5, "белиберда без слова на Р"))

documents = [

```

```

Document(1, 1, "Документ 1", 100),
Document(2, 1, "Документ 2", 200),
Document(3, 2, "Документ 3", 300),
Document(4, 3, "Документ 4", 400),
]

doc_sections = [
    DocSection(1, 1),
    DocSection(1, 2),
    DocSection(2, 3),
    DocSection(3, 4),
]

def test_task1():
    result = task1(documents, sections)
    assert result[0][0] == "Раздел 1"
    assert result[0][1] == ["Документ 1", "Документ 2"]

def test_task2():
    result = task2(documents, sections)
    assert result[0] == ("Раздел 3", 400)

def test_task3():
    result = task3(documents, sections, doc_sections)
    assert ("белиберда без слова на Р", []) not in result

```

Результат тестов:

```

(venv) sans2114@WIN-C4NK5RA1UES:~/Schebletsov_Lev_3Sem_labs$ pytest
===== test session starts =====
platform linux -- Python 3.12.3, pytest-9.0.2, pluggy-1.6.0
rootdir: /home/sans2114/Schebletsov_Lev_3Sem_labs
collected 3 items

test_main.py ... [100%]

===== 3 passed in 0.02s =====
NO NEW NOTIFICATIONS
(venv) sans2114@WIN-C4NK5RA1UES:~/Schebletsov_Lev_3Sem_labs$

```